Hydraulic System Troubleshooting Guide

Synthetic project reference document for Phase 5 RAG testing.

1. Hydraulic Monitoring

Hydraulic pressure is an important indicator for machines using hydraulic clamping, actuation, or power systems. Pressure should be interpreted against the machine's approved operating range.

2. Low Pressure

Potential causes include fluid leakage, low fluid level, clogged filters, pump wear, relief-valve problems, air ingress, or excessive internal leakage. Check fluid level, visible leaks, filter condition, pump behavior, and pressure trends.

3. Pressure Instability

Fluctuating pressure can result from air in the system, unstable pump operation, valve problems, contamination, or changing load. Correlate pressure behavior with machine operation and recent maintenance.

4. Preventive Maintenance

Maintain fluid cleanliness, inspect hoses and fittings, monitor filters, check fluid level and condition, and record pressure measurements during routine maintenance. Hydraulic leaks require site-approved safety handling.

5. Troubleshooting Sequence

Verify safe isolation before physical inspection. Confirm pressure, inspect fluid level and leaks, check filter condition, inspect pump and valve behavior, and compare the condition with historical maintenance records.